

# Untrained CNNs Exceed Backpropagation in V1 Alignment at High Evaluation Resolution: A Systematic RSA Comparison of Four Learning Rules Against Human fMRI

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## Abstract

**Revised version (v4).** This version replaces the noise ceiling, restricts all analyses to stimulus pairs that were presented in different fMRI runs, and uses the repaired evaluation of all five conditions; see the revision note below.

A central question in computational neuroscience is whether the learning rule used to train a neural network determines how well its internal representations align with those of the human visual cortex. We present a systematic comparison of four learning rules—backpropagation (BP), feedback alignment (FA), predictive coding (PC), and spike-timing-dependent plasticity (STDP)—applied to identical convolutional architectures and evaluated against human fMRI data from the THINGS-fMRI dataset (720 stimuli, 3 subjects) using Representational Similarity Analysis (RSA). All models process stimuli at  $224 \times 224$  resolution; results are averaged across 5 random seeds, and uncertainty is estimated by resampling stimuli. Crucially, we include an untrained random-weights baseline that isolates the contribution of architecture. At V1, the untrained baseline *exceeds* backpropagation ( $\rho = 0.053$  vs.  $\rho = 0.022$ ;  $\Delta\rho = +0.031$ , 95% CI [0.019, 0.043]), and the same holds at V2. This advantage depends on the evaluation resolution: the models are trained at 32 px and evaluated at 224 px. In the 5-seed model set of the resolution study (same architecture and training; gap +0.030 at 224 px), the gap vanishes when the models are evaluated at 32 px ( $\Delta\rho = -0.001$ , 95% CI [-0.011, 0.009]; arXiv:2608.12408). At LOC, BP, PC and STDP exceed the random baseline, FA is borderline, and the trained rules do not differ from each other. At IT, no two conditions can be distinguished; the reliability of the IT data is low (leave-one-subject-out lower bound 0.021). FA produces the lowest alignment at V1. Only the untrained network reaches the lower noise bound at V1; at V2 it and STDP are not distinguishable from the bound, and every condition lies below it at LOC and IT. These results indicate that, at the 224 px evaluation resolution used here, early visual alignment is driven more by the architecture than by the learning rule,

and that learning-rule differences are resolvable only at intermediate areas.

**Keywords:** representational similarity analysis, visual cortex, biologically plausible learning, random baseline, STDP, feedback alignment, predictive coding, fMRI, THINGS

## Revision note (v4)

Earlier versions contained the following errors. (i) The noise ceiling was described as split-half reliability with a Spearman–Brown correction; with three subjects that is not a valid estimate, and the printed bounds could not be reproduced from any stored result. It is replaced by the leave-one-subject-out lower bound of Nili et al. (2014); the upper bound is not informative at  $N = 3$  and is reported only in the appendix. (ii) Subjects sub-01 and sub-02 saw the stimuli in identical order, so pairs of stimuli from the same run carry a run component that is common to these two subjects; it inflated their agreement and the lower bound at LOC and IT. All analyses now use only the 210,205 of 258,840 stimulus pairs (81.2%) that lie in different runs for every subject. (iii) The PC and STDP conditions were evaluated with batch normalisation left in training mode. The correction note of v3 (August 2026) identified this defect, but the values reported in the paper were not updated; all values are now from the repaired five-seed re-run. (iv) Several values were rounded twice (e.g. 0.075 printed as 0.076), and the V1 gap was quoted from a different run (appendix, Table 12). (v) Two statements about LOC contradicted each other. (vi) The hierarchy figure showed, for each layer, the mean over all four ROIs, contrary to its caption, which described the fixed layer-to-ROI mapping; it now shows that mapping. (vii) The untrained network was described as He-normal initialised; it uses PyTorch’s default `kaiming_uniform_`, while FA and STDP use `kaiming_normal_`. (viii) The accuracy table reported accuracy on training batches; it now reports CIFAR-10 test accuracy. What changes: the lower bounds are 0.051, 0.024, 0.018 and 0.021 (V1 to IT);

the claim that the models “exhaust most of the available signal” is withdrawn; PC no longer exceeds BP at V1, and STDP exceeds it only in the per-subject convention; IT results are stated as not distinguishable at low reliability; statistics are stimulus-bootstrap CIs instead of permutation tests. What holds: the untrained network exceeds BP at V1 and V2, BP exceeds the untrained network at LOC, and FA has the lowest alignment at V1. FA at V2 is no longer a primary comparison, because its ordering against BP and PC depends on the layer assigned to V2 (appendix, Table 6); reducing the family of primary comparisons from 11 to 7 does not change any significance decision (appendix, Table 7). Still on the original state are the filter and Gabor analysis (weights of the original model set; unaffected by the defect), the training curves, and the batch-normalisation calibration control (all stimulus pairs, mean-RDM convention); test accuracy is available for seed 0 only.

## 1 Introduction

Deep neural networks (DNNs) trained with backpropagation have become the dominant computational models of the ventral visual stream (Yamins and DiCarlo, 2016). A well-established finding is that the hierarchy of features learned by DNNs mirrors the hierarchy of representations from V1 through IT cortex, as quantified by representational similarity analysis (RSA) and related metrics (Kriegeskorte et al., 2008; Schrimpf et al., 2020). This correspondence has been interpreted as evidence that the objectives and inductive biases of deep learning capture something fundamental about visual processing.

Yet backpropagation is neurobiologically implausible in at least three well-known respects: it requires symmetric forward and feedback weights (the weight transport problem), it relies on a global error signal propagated backward through the network, and the forward and backward passes must be temporally separated (Crick, 1989; Lillicrap et al., 2020). These observations have motivated work on biologically plausible alternatives, including feedback alignment (Lillicrap et al., 2016), predictive coding (Rao and Ballard, 1999; Whittington and Bogacz, 2017), and spike-timing-dependent plasticity (Bi and Poo, 1998).

A key question is: does the choice of learning rule matter for brain alignment? If alignment arises from the architecture and objective rather than the weight-update rule, then alternative rules—or even no training at all—should yield comparable alignment. Prior work has compared individual learning rules to brain data (Schrimpf et al., 2020), but to our knowledge no study has systematically compared all four rules together with an untrained random-weights baseline on the same architecture, stimuli, and RSA pipeline.

We address this gap with five conditions: BP, FA, PC,

STDP, and a random-weights (untrained) baseline, all applied to identical CNNs trained on CIFAR-10, with layer-wise RDMs compared to human fMRI from the THINGS-fMRI dataset.

Our main contributions are: (1) a head-to-head RSA comparison of four learning rules plus a random baseline at  $224 \times 224$  resolution, averaged across 5 seeds; (2) the finding that, at 224px evaluation, the untrained CNN exceeds BP at V1 and V2 (V1:  $\Delta\rho = +0.031$ , 95% CI [0.019, 0.043]), while at V1 the two do not differ at the 32px training resolution; (3) the result that at IT, no two conditions can be distinguished at the low reliability of the IT data; (4) the finding that LOC is the only area where trained rules exceed the untrained baseline: BP, PC and STDP do so while the trained rules do not differ from each other; and (5) the finding that FA produces the lowest brain alignment at V1.

## 2 Related Work

**RSA as a model evaluation tool.** Kriegeskorte et al. (2008) introduced RSA as a framework for comparing representations across measurement modalities. The core idea is to represent each system’s response to a stimulus set as a pairwise dissimilarity matrix (RDM) and to assess model–brain alignment as the rank correlation between RDMs. RSA has since become a standard tool for comparing DNNs to neural data (Yamins and DiCarlo, 2016; Schrimpf et al., 2020).

**DNNs and the visual hierarchy.** Task-optimized DNNs reproduce the representational hierarchy of the ventral visual stream, with early layers matching V1 and deeper layers matching IT (Yamins and DiCarlo, 2016). The Brain-Score benchmark (Schrimpf et al., 2020) systematizes this evaluation across architectures and datasets. Most high-scoring models are trained with standard BP on ImageNet. Importantly, Schrimpf et al. (2020) noted that untrained networks retain some brain alignment, but did not systematically compare this against multiple learning rules.

**Biologically plausible learning rules.** Feedback alignment (Lillicrap et al., 2016) replaces the transpose weight matrix in the backward pass with fixed random matrices, removing the weight transport problem. Lillicrap et al. (2016) showed that FA still allows learning in fully-connected networks, though its effectiveness in convolutional architectures is less established. Predictive coding (Rao and Ballard, 1999) frames inference as hierarchical prediction-error minimization; Whittington and Bogacz (2017) showed that PC implements an approximation to BP under certain conditions, with learning driven entirely by local Hebbian updates. Spike-timing-dependent plasticity (Bi and Poo, 1998) updates synaptic weights based on the relative timing of pre- and postsynaptic spikes, and is one of the best-characterized forms of synaptic plasticity in mammalian cortex. Each of these

rules has been studied individually in the context of neural representation learning, but direct comparisons on identical architectures and evaluation pipelines are rare.

**Random networks and early visual cortex.** Saxe et al. (2011) demonstrated that random-weight networks can exhibit non-trivial visual structure. Concurrently with our work, Truzzi and Cusack (2025) showed that random-weight CNNs predict early visual cortex responses when model complexity is sufficient. Our work extends these observations by systematically comparing four distinct learning rules against the random baseline across the visual hierarchy.

**Sparse coding and V1.** Olshausen and Field (1996) showed that unsupervised learning of a sparse code for natural images produces receptive fields resembling V1 simple cells, establishing that correlation-based learning from image statistics suffices to produce V1-like representations without supervised objectives.

## 3 Methods

### 3.1 Dataset

We used the THINGS-fMRI dataset, part of THINGS-data (Hebart et al., 2023): 3T fMRI responses (Siemens Magnetom Prisma) of 3 subjects to 8,740 unique images of 720 object concepts from the THINGS database (Hebart et al., 2019), recorded in 12 main-experiment sessions of 10 runs each. We selected 720 stimuli (one image per concept, each presented once). fMRI responses were available for 3 subjects across 4 regions of interest (ROIs): V1, V2, LOC, and IT. For each subject and ROI, single-trial responses were z-scored per voxel across all trials, and a subject RDM was computed as the correlation distance between the response patterns of the 720 stimuli.

### 3.2 Model Architecture

All five conditions used an identical CNN architecture consisting of three convolutional layers (Conv1–Conv3, each followed by BatchNorm, ReLU, and  $2 \times 2$  max-pooling) and two fully connected layers (FC1, FC2). The output layer uses softmax for 10-class classification on CIFAR-10. Training used 8,000 CIFAR-10 samples for 40 epochs at native  $32 \times 32$  resolution. For RSA evaluation, THINGS stimuli were resized to  $224 \times 224$  and passed through the same trained (or untrained) network; this is standard practice in model–brain comparisons where training and evaluation datasets differ (Schrumpf et al., 2020). All results are from the five-seed re-run with the repaired evaluation mode (Revision note, item iii); the original model set is reported as a sensitivity column in the appendix.

The fixed layer-to-ROI mapping is Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT. V1 and V2 are

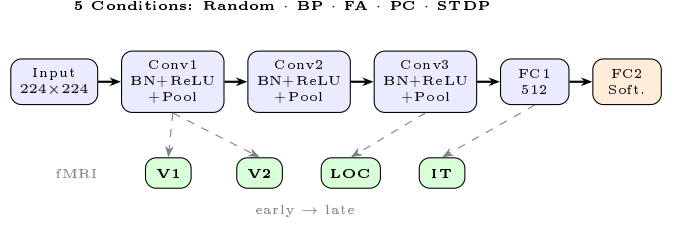


Figure 1: Model architecture and layer-to-ROI mapping. Models are trained on CIFAR-10 ( $32 \times 32$ ); for RSA, THINGS stimuli are resized to  $224 \times 224$  at feature extraction time. Dashed arrows: Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT.

both mapped to Conv1; Conv2 is not used in the fixed mapping (its values are given in the best-layer analysis, Table 3).

### 3.3 Learning Rules

**Random Weights** serves as the architectural baseline. The network is initialized with PyTorch’s default `kaiming_uniform_` and never trained, isolating the contribution of the convolutional architecture (filters, ReLU, pooling) to brain alignment. FA and STDP initialize with `kaiming_normal_`, whose standard deviation is larger by a factor of about 2.4. The untrained network therefore shares its initialization with BP and PC and is their matched control; for FA and STDP, comparisons with the untrained network also include the difference in initialization scale.

**Backpropagation (BP)** serves as the supervised baseline:  $\Delta W_l = -\eta \frac{\partial \mathcal{L}}{\partial W_l}$ .

**Feedback Alignment (FA)** (Lillicrap et al., 2016) replaces  $W_l^\top$  in the backward pass with a fixed random matrix  $B_l$ :

$$\Delta W_l = \delta_l^{\text{FA}} \mathbf{h}_{l-1}^\top, \quad \delta_l^{\text{FA}} = f'(z_l) \odot (B_{l+1} \delta_{l+1}^{\text{FA}}). \quad (1)$$

**Predictive Coding (PC)** (Rao and Ballard, 1999; Whittington and Bogacz, 2017) minimizes a hierarchical prediction-error energy  $F = \sum_l \|\varepsilon_l\|^2$  through iterative inference ( $T_{\text{inf}} = 10$  steps, inference rate  $\alpha = 0.02$ ):

$$\Delta W_l \propto \varepsilon_l \mathbf{x}_{l-1}^\top. \quad (2)$$

Prediction weights  $\mathbf{P}_l$  (implemented as transposed convolutions) generate top-down predictions; prediction errors  $\varepsilon_l = r_l - \hat{r}_l$  drive both inference dynamics (updating representations  $r_l$ ) and weight learning (weight rate  $\eta_W = 10^{-4}$ ). As with STDP, only convolutional weights are updated via PC; the fully connected readout layers are trained with backpropagation.

**STDP** (Bi and Poo, 1998) converts activations to Poisson spike trains ( $T = 10$  timesteps) and uses first-

spike timing to update convolutional weights:

$$\Delta w = \begin{cases} A_+ e^{-\Delta t/\tau_+} & \Delta t > 0 \text{ (LTP)} \\ -A_- e^{\Delta t/\tau_-} & \Delta t < 0 \text{ (LTD)} \end{cases} \quad (3)$$

with  $\tau_+ = \tau_- = 20$  ms,  $A_+ = A_- = 0.003$ ,  $\text{lr} = 5 \times 10^{-4}$ . Only the convolutional layers (Conv1–Conv3) are updated via STDP; the fully connected layers (FC1, FC2) are trained with backpropagation to provide a supervised readout, analogous to a linear probe on STDP-learned features. The STDP test accuracy (57.4%) therefore reflects STDP feature quality evaluated through a BP-trained classifier, not end-to-end STDP learning.

### 3.4 RSA Pipeline

**Feature extraction.** For each layer, activations were extracted on the 720 THINGS stimuli (resized to  $224 \times 224$ , standard ResNet input resolution). Convolutional activations were spatially averaged (global average pooling). All results are averaged across 5 random seeds per learning rule.

**RDM construction.** Model RDMs were computed as  $1 - r$  (correlation distance).

**RSA score.** Model–brain alignment is Spearman’s  $\rho$  between the upper triangles of a model RDM and a subject RDM, restricted to the stimulus pairs defined below. The main convention ( $\rho_{\text{subj}}$ ) correlates the model RDM with each subject’s RDM and averages over the three subjects. The convention of earlier versions ( $\rho_{\text{mean}}$ ) correlates the model RDM with the average of the three subject RDMs; it is reported alongside.

**Stimulus pairs (cross-run).** Each subject saw the 720 stimuli in ten runs, and sub-01 and sub-02 saw them in identical order. Responses within a run can share run-specific components (baseline drifts, run-wise noise), so for two subjects with the same order the dissimilarities of same-run pairs share a component that has nothing to do with the stimuli. This inflated the agreement between sub-01 and sub-02 (at LOC 0.062 on all pairs vs. 0.023 on cross-run pairs, while sub-01 vs. sub-03 is 0.008 and 0.009) and thereby the lower bound at LOC and IT (appendix, Table 5). We therefore use only stimulus pairs that lie in different runs for all three subjects: 210,205 of 258,840 pairs; 18.8% are excluded. Within a run, dissimilarity also depends on the distance between trials (Spearman correlation between dissimilarity and lag  $-0.30$  to  $-0.10$ ); at LOC and IT, neighbouring trials are on average anticorrelated (mean dissimilarity at lag 1 up to 1.030). This would be consistent with collinearity between the overlapping hemodynamic responses of neighbouring trials in single-trial estimates, which is a further reason to exclude these pairs; we did not test this explanation. Excluding same-run pairs follows the same logic as run-wise cross-validation of dissimilarities, which keeps noise shared within a run out of the dissimilarity estimates (Walther et al., 2016).

**Noise ceiling.** The lower bound follows Nili et al. (2014): each subject’s RDM is correlated with the average RDM of the other two subjects, and the three correlations are averaged. It estimates how well a model that captures the structure shared across subjects would score, and is used here as a floor: a model below it leaves reliable shared structure unexplained. The upper bound (correlation of each subject with the average of all three, including itself) is not informative with three subjects: under a permutation null in which stimulus labels are shuffled independently per subject (1000 permutations), the upper bound stays large (V1: 0.548 against an observed 0.572), because each subject contributes a third of the average it is compared with. It is therefore not shown in the figures. Under the same null the lower bound is  $0.000 \pm 0.002$  (V1), so the observed lower bounds are far from chance at every ROI.

**Uncertainty.** 95% confidence intervals come from a stimulus bootstrap: 1000 resamples of the 720 stimuli with replacement (seed 20261002), shared by all conditions, ROIs and quantities. Within each resample, pairs of a stimulus with itself are dropped and the cross-run pair set is recomputed. Differences between conditions, and between a condition and the lower bound, are computed on the same resamples, so their intervals account for the correlation between them. We call a difference significant when its percentile interval excludes 0. The three statements the conclusions rest on are treated as primary comparisons: the untrained network vs. BP at V1 and V2, BP vs. the untrained network at LOC, and FA vs. each other condition at V1 (7 of the 40 pairwise comparisons). They were designated in this revision, not before the original analysis. All other pairwise comparisons are exploratory. No correction for multiple testing is applied in the main text; all 40 intervals are reported, with the primary ones marked (appendix, Table 10), and Benjamini–Hochberg and Holm corrections of the primary family are given in the appendix (Table 7).

### 3.5 Partial RSA

To control for low-level pixel similarity, a pixel RDM is computed as the pairwise correlation distance between the flattened RGB images (resized to 224px on the shorter edge and centre-cropped to  $224 \times 224$ , as for the networks). Partial Spearman  $\rho$  is estimated by residualizing the ranks of the model and the brain RDM vectors on the ranks of the pixel RDM vector and correlating the residuals. The pixel RDM was rebuilt for this version from the definition in the code of earlier versions; applied to the earlier analysis it reproduces the stored values to within 0.00005 (they were stored at four decimals). Partial  $\rho$  is computed per seed and subject on cross-run pairs and averaged as  $\rho_{\text{subj}}$ , with the same stimulus bootstrap.

## 4 Results

### 4.1 Task Performance

Table 1: CIFAR-10 test accuracy (10,000 test images) of the repaired models after 40 epochs. Seed 0 only: checkpoints of the other four seeds were not stored, so no standard deviation can be given. Earlier versions reported training accuracy (appendix, Table 11).

| Condition          | Test accuracy (%) |
|--------------------|-------------------|
| Backprop           | 75.0              |
| STDP               | 57.4              |
| Predictive Coding  | 56.7              |
| Feedback Alignment | 38.8              |
| Random Weights     | 10.0              |

### 4.2 Layer-wise Brain Alignment

**Interpreting absolute RSA scores.** Absolute Spearman  $\rho$  values in this study are low (e.g.,  $\rho < 0.10$  at V1). They are not comparable with the scores of encoding-model benchmarks: RSA and encoding models measure fundamentally different quantities: RSA evaluates the geometry of the representational space (pairwise dissimilarity structure), whereas encoding models predict individual voxel responses, which is a more permissive criterion that can exploit stimulus-specific variance. RSA scores are further attenuated by the spatial resolution of fMRI relative to electrophysiology, and by our relatively small CNN architecture. The data themselves are of limited reliability: the lower bound is 0.051 [0.039, 0.064] at V1, 0.024 at V2, 0.018 at LOC and 0.021 at IT. Taken as a floor, it separates the conditions (model – lower bound, with 95% CI). At V1, only the untrained network reaches it (+0.002 [−0.014, 0.017]); every trained rule lies below it. At V2, the untrained network (+0.005 [−0.007, 0.017]) and STDP (−0.009 [−0.021, 0.002]) are not distinguishable from it, BP lies just below it (its interval ends less than  $10^{-4}$  below zero), and FA and PC lie below it. At LOC and IT every condition lies below it. Reaching the lower bound is a weak criterion at V1: a single luminance value per image reaches  $\rho_{\text{subj}} = 0.054$  against the same V1 data, as much as the untrained network (0.053; arXiv:2608.12408).

Table 2 provides the full numerical results. Several key patterns emerge.

**V1/V2—Untrained exceeds BP at 224px evaluation.** In early visual cortex, the untrained random-weights baseline *exceeds* BP at V1 ( $\rho = 0.053$  vs.  $\rho = 0.022$ ;  $\Delta\rho = +0.031$  [0.019, 0.043]) and at V2 (0.029 vs. 0.012;  $\Delta\rho = +0.017$  [0.007, 0.026]). The order at V1 is Random > STDP > BP > PC > FA, and all ten pairwise differences at V1 are significant in the per-subject convention; in the mean-RDM convention BP

and STDP do not differ ([−0.0113, 0.0005]), and at V2 they do not differ in either convention. At this evaluation resolution, the convolutional architecture—local connectivity, ReLU, and pooling—is thus the primary driver of early visual alignment. When the models are evaluated at their 32px training resolution, BP and the untrained network do not differ at V1 ( $\rho_{\text{subj}} = 0.046$  vs. 0.045;  $\Delta\rho = -0.0007$ , 95% CI [−0.0109, +0.0090]; cross-run pairs, model set of the resolution study, same architecture and training, 5 seeds); the untrained network’s advantage at 224px arises from the difference between training and evaluation resolution (arXiv:2608.12408).

**LOC—Learning effect without differences among rules.** At LOC, BP (0.008), PC (0.004) and STDP (0.004) exceed the random baseline (−0.003; BP – Random +0.011 [0.002, 0.021], PC – Random +0.007 [0.001, 0.014], STDP – Random +0.007 [0.001, 0.014]). FA is borderline: its difference from Random is +0.0066 [−0.0002, 0.0140] in the per-subject convention and excludes 0 in the mean-RDM convention [0.002, 0.026]. No difference between two trained rules is significant (e.g. BP – FA +0.004 [−0.001, 0.010]).

**IT—Not distinguishable at low reliability.** At IT, the five conditions range from 0.004 to 0.009, and none of the ten pairwise differences is significant, including Random vs. BP (−0.005 [−0.014, 0.004]). Reliability at IT is low, second lowest after LOC (lower bound 0.021 [0.014, 0.028]), and every condition lies below this bound. The conditions are therefore not distinguishable at the reliability available at IT; this does not show that they are equivalent.

**FA lowest at V1.** FA produces the lowest brain alignment at V1 (0.009), and its differences from every other condition at V1 are significant in both conventions, suggesting that random feedback filters disrupt the representational structure that the convolutional architecture provides for free. This holds despite FA reaching 38.8% test accuracy—substantially above chance—indicating that FA learns task-relevant features that do not correspond to human visual representations. At V2, the ordering among FA, BP and PC depends on the layer assignment (Conv1 vs. Conv2; Appendix Table 6), whereas Random > BP holds under both.

### 4.3 Hierarchical Gradient

All five conditions are lower at LOC than at V1 (Figure 4). The random baseline shows the most distinctive pattern: highest at V1/V2 (exceeding all trained rules), dropping below zero at LOC (−0.003, the lowest), and at IT (0.004) the lowest again but not significantly different from any trained rule. Convolutional inductive biases thus drive early alignment, while at IT the differences between conditions are below what the data can resolve.

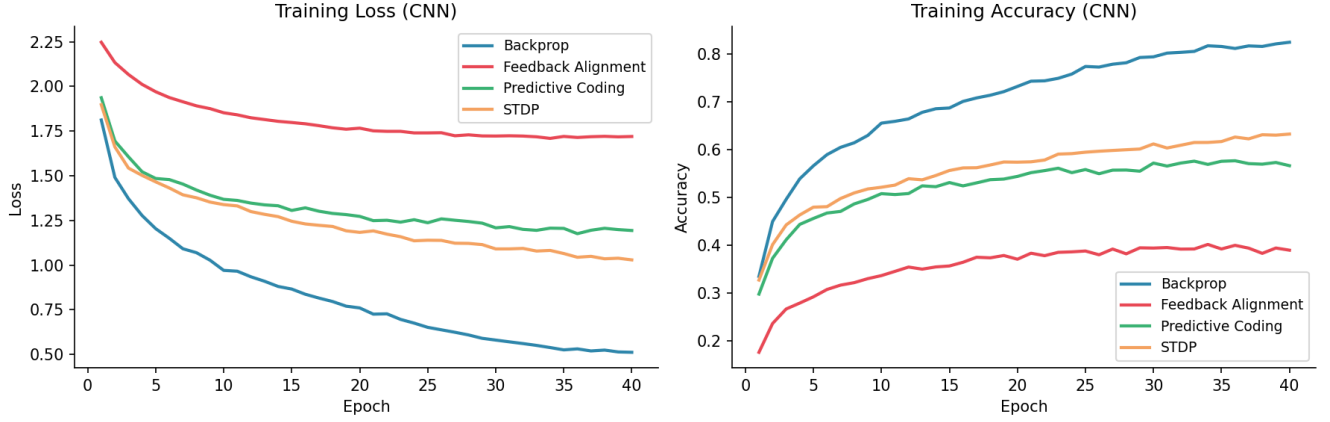


Figure 2: Training dynamics of the original runs: loss (left) and accuracy (right) on the training batches over 40 epochs (8,000 CIFAR-10 samples) for all four learning rules. All runs are stable without divergence. Test accuracies of the repaired models are in Table 1.

Table 2: RSA scores  $\rho_{\text{subj}}$  (per-subject convention; mean across 5 seeds) with stimulus-bootstrap 95% CI, cross-run stimulus pairs, repaired model set,  $224 \times 224$  input. The mean-RDM convention is in the appendix (Table 9). Lower bound: Nili et al. (2014) leave-one-subject-out. Bold: highest  $\rho_{\text{subj}}$  per ROI (V1, V2, LOC).  $^{\dagger}$ : difference from Random Weights significant (CI excludes 0).  $^{\ddagger}$ : at IT no pairwise difference is significant; no condition is marked.

| ROI | Layer | lower bound        | Random                    | BP                                     | FA                               | PC                               | STDP                             |
|-----|-------|--------------------|---------------------------|----------------------------------------|----------------------------------|----------------------------------|----------------------------------|
| V1  | Conv1 | 0.051 [.039, .064] | <b>0.053</b> [.039, .069] | 0.022 [.013, .033] $^{\dagger}$        | 0.009 [.000, .018] $^{\dagger}$  | 0.012 [.003, .022] $^{\dagger}$  | 0.027 [.017, .037] $^{\dagger}$  |
| V2  | Conv1 | 0.024 [.016, .032] | <b>0.029</b> [.019, .040] | 0.012 [.005, .020] $^{\dagger}$        | 0.004 [−.003, .011] $^{\dagger}$ | 0.006 [−.001, .013] $^{\dagger}$ | 0.015 [.007, .023] $^{\dagger}$  |
| LOC | Conv3 | 0.018 [.011, .027] | −0.003 [−.010, .004]      | <b>0.008</b> [.001, .015] $^{\dagger}$ | 0.004 [−.004, .011]              | 0.004 [−.002, .011] $^{\dagger}$ | 0.004 [−.002, .010] $^{\dagger}$ |
| IT  | FC1   | 0.021 [.014, .028] | 0.004 [−.002, .010]       | 0.009 [.004, .015]                     | 0.008 [.001, .014]               | 0.008 [.002, .013]               | 0.009 [.003, .014] $^{\ddagger}$ |

#### 4.4 Pairwise Comparisons

All 40 pairwise differences (10 pairs  $\times$  4 ROIs) with their bootstrap intervals are listed in Table 10. At V1, all ten are significant; at V2, all except BP vs. STDP. At LOC, the differences between the random baseline and BP, PC and STDP are significant, Random vs. FA is borderline, and none of the six differences among trained rules is significant. At IT, none of the ten is significant.

#### 4.5 Subject-Level Consistency

We assessed whether the results hold within individual subjects (Figure 5; single-subject  $\rho$  with stimulus-bootstrap CIs on the same resamples). Sub-03 shows weaker alignment across all conditions and ROIs, suggesting lower fMRI signal quality. At V1, sub-01 and sub-02 each show the group order Random  $>$  STDP  $>$  BP  $>$  PC  $>$  FA in their point estimates, and in each of them the untrained network exceeds every trained rule (e.g. Random  $-$  BP: sub-01  $+0.033$  [0.016, 0.051], sub-02  $+0.052$  [0.026, 0.076]). FA is below every other condition in sub-01; in sub-02 its difference from PC is borderline. At LOC, single subjects do not resolve the differences: BP has the highest point estimate in sub-02 and sub-03, and in sub-01 STDP (0.007) is marginally above BP (0.006); no difference between two trained rules is sig-

nificant in any subject, and the difference between the untrained network and BP is significant only in sub-02 (Random  $-$  BP  $-0.024$  [−0.047, −0.001]). The LOC result of the main analysis therefore rests on the combination of subjects, not on a consistent per-subject ranking.

#### 4.6 Seed Variability

To assess robustness to random initialization, we trained each learning rule with 5 different random seeds (Figure 6). At V1/V2, the seed standard deviation (0.001–0.006) is small relative to the differences between conditions (e.g.  $\Delta\rho = +0.031$  between Random and BP at V1), so the order of conditions is stable across initializations. At IT, the seed standard deviations are also small (0.0005–0.0020), and most seed ranges overlap; BP’s lowest seed (0.00865) lies just above FA’s highest (0.00862) and above the untrained network’s highest (0.00725). The seed spread does not include the uncertainty from the sampling of stimuli, which is much larger (Random  $-$  BP at IT: [−0.014, 0.004]); it is this uncertainty that leaves the IT comparisons unresolved.

#### 4.7 Best-Layer-per-ROI Analysis

We evaluated whether the fixed layer-to-ROI mapping is optimal by computing RSA for all layer $\times$ ROI combi-

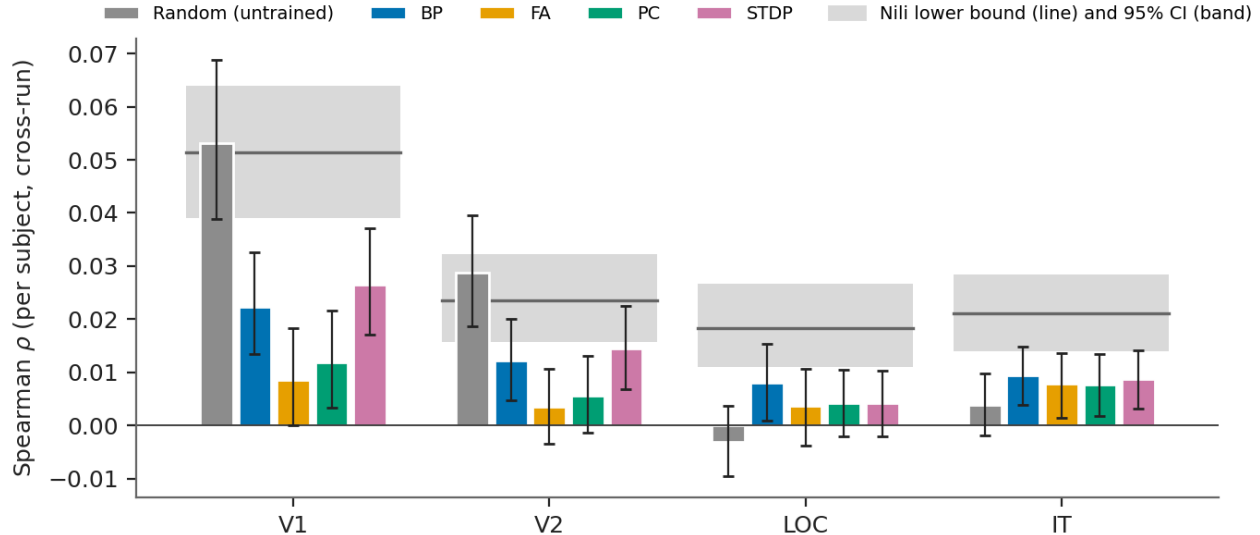


Figure 3: Brain alignment across ROIs (main result). Spearman  $\rho$  between model RDMs and each subject’s RDM, averaged over the 3 subjects ( $\rho_{\text{subj}}$ ), cross-run stimulus pairs, mean of 5 seeds. Error bars: stimulus-bootstrap 95% CI. Grey line and band: Nili et al. (2014) lower bound and its 95% CI. The upper bound is not shown because it is not informative with three subjects (Methods).

nations (Table 3; point estimates,  $\rho_{\text{subj}}$ , cross-run). For BP, the fixed mapping matches the best layer at 3 of four ROIs; at LOC, FC1 is higher than Conv3. For Random, FA and PC, FC1 produces the highest V1 alignment (Random: 0.102 vs. 0.053 at Conv1)—likely reflecting that FC1 compresses all information into a 512-dimensional space that happens to correlate with the fMRI signal. This finding suggests that best-layer analysis can be misleading when layer dimensionality varies, and supports the use of fixed anatomically-motivated mappings.

Table 3: Best layer per ROI and its  $\rho_{\text{subj}}$  (cross-run, mean of 5 seeds; point estimates). In parentheses: the fixed layer’s value where it differs.

| Condition | V1                  | V2                | LOC                   | IT        |
|-----------|---------------------|-------------------|-----------------------|-----------|
| Random    | FC1 0.102 (0.053)   | FC1 0.061 (0.029) | Conv1 -0.002 (-0.003) | FC1 0.004 |
| BP        | Conv1 0.022         | Conv1 0.012       | FC1 0.012 (0.008)     | FC1 0.009 |
| FA        | FC1 0.020 (0.009)   | FC1 0.014 (0.004) | FC1 0.005 (0.004)     | FC1 0.008 |
| PC        | FC1 0.015 (0.012)   | FC1 0.013 (0.006) | Conv2 0.005 (0.004)   | FC1 0.008 |
| STDP      | Conv2 0.027 (0.027) | FC1 0.018 (0.015) | FC1 0.007 (0.004)     | FC1 0.009 |

## 4.8 Partial RSA

Removing the variance explained by pixel-level similarity lowers every condition’s V1 score ( $\Delta$  between  $-0.005$  and  $-0.002$ ), and each of these decreases is significant (Table 4). The order at V1 is unchanged (Random > STDP > BP > PC > FA), and the untrained network still exceeds BP at V1 ( $+0.034$  [0.021, 0.046]) and V2 ( $+0.018$  [0.008, 0.027]). Across all 40 pairwise comparisons, the same differences are significant with and without pixel

control, so none of the results reported above is carried by pixel-level similarity.

## 4.9 Filter Analysis

Conv1 filters from all five conditions exhibit color-opponent patch structure rather than oriented Gabor-like patterns, consistent with CIFAR-10 training. The Gabor-peakedness score (FFT spectral peak-to-mean ratio) is highest for BP ( $1.99 \pm 0.52$ ), followed by FA ( $1.82 \pm 0.36$ ), STDP ( $1.79 \pm 0.47$ ), PC ( $1.73 \pm 0.45$ ), and Random ( $1.73 \pm 0.45$ ). The differences are small relative to within-rule variability and standard deviations overlap substantially across all conditions, indicating that no learning rule produces markedly more Gabor-like filters than random initialization at this scale.

## 5 Discussion

**Architecture as the primary driver of early alignment.** Our most striking finding is that an untrained CNN *exceeds* backpropagation at V1 ( $\rho = 0.053$  vs. 0.022,  $\Delta\rho = +0.031$  [0.019, 0.043]). This goes beyond the observation of Saxe et al. (2011) that random networks have non-trivial visual structure: in our setup, at 224 px evaluation, the convolutional inductive biases (local connectivity, ReLU, pooling) are not merely comparable to BP but produce higher early visual alignment. The advantage is not evidence that training degrades early-layer structure: when the models are evaluated at their 32 px training resolution, BP and the untrained network do

Table 4: Standard ( $\rho_{\text{subj}}$ ) and partial RSA ( $\rho_{\text{partial}}$ , pixel similarity controlled) at V1 and V2;  $\Delta = \rho_{\text{partial}} - \rho_{\text{subj}}$ . Cross-run pairs, repaired model set, per-subject convention, mean of 5 seeds, stimulus-bootstrap 95% CI.

| Condition | V1                   |                                  |                            | V2                   |                                  |                            |
|-----------|----------------------|----------------------------------|----------------------------|----------------------|----------------------------------|----------------------------|
|           | $\rho_{\text{subj}}$ | $\rho_{\text{partial}}$ [95% CI] | $\Delta$ [95% CI]          | $\rho_{\text{subj}}$ | $\rho_{\text{partial}}$ [95% CI] | $\Delta$ [95% CI]          |
| Random    | 0.053                | 0.051 [0.037, 0.066]             | -0.0025 [-0.0042, -0.0011] | 0.029                | 0.028 [0.018, 0.038]             | -0.0011 [-0.0024, -0.0000] |
| STDP      | 0.027                | 0.022 [0.013, 0.032]             | -0.0045 [-0.0069, -0.0024] | 0.015                | 0.013 [0.005, 0.020]             | -0.0020 [-0.0041, -0.0002] |
| BP        | 0.022                | 0.017 [0.009, 0.027]             | -0.0052 [-0.0080, -0.0027] | 0.012                | 0.010 [0.003, 0.017]             | -0.0022 [-0.0045, -0.0002] |
| PC        | 0.012                | 0.007 [-0.001, 0.017]            | -0.0046 [-0.0071, -0.0026] | 0.006                | 0.004 [-0.003, 0.011]            | -0.0021 [-0.0041, -0.0005] |
| FA        | 0.009                | 0.004 [-0.004, 0.013]            | -0.0045 [-0.0069, -0.0024] | 0.004                | 0.002 [-0.005, 0.008]            | -0.0020 [-0.0040, -0.0003] |

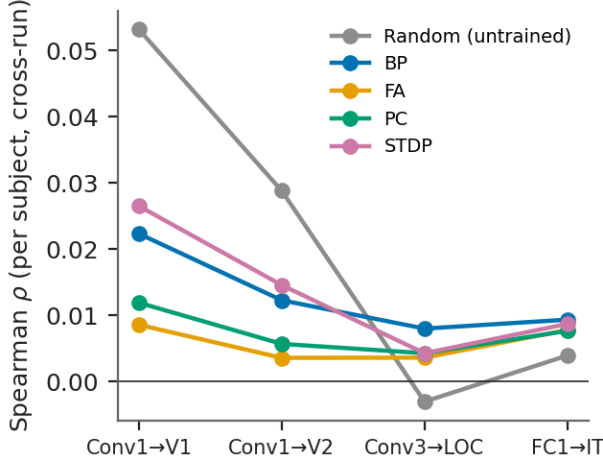


Figure 4: Brain alignment along the fixed layer-to-ROI mapping ( $\rho_{\text{subj}}$ , cross-run). All conditions are lower at LOC than at V1. At IT the conditions lie between 0.004 and 0.009 and are not distinguishable. The untrained network is the highest at V1/V2 and the lowest at LOC and IT.

not differ at V1 ( $\rho_{\text{subj}} = 0.046$  vs.  $0.045$ ;  $\Delta\rho = -0.0007$ , 95% CI  $[-0.0109, +0.0090]$ , model set of the resolution study), and the untrained network’s lead at 224px arises from the difference between training and evaluation resolution (arXiv:2608.12408). Whether the same holds at larger scales (e.g., ResNet-50 on ImageNet) remains to be tested.

**On the untrained baseline.** The untrained condition’s batch-normalisation layers hold their initialised statistics, so in evaluation mode they act as the identity, whereas every trained condition carries statistics accumulated during training. A  $2 \times 2$  calibration control varying the calibration set and the calibration resolution, with the convolutional weights held bit-identical, indicates that this does not produce the result: with statistics calibrated at the evaluation resolution the untrained network still exceeds backpropagation at V1 by  $+0.041$  (CIFAR-10 statistics) or  $+0.033$  (statistics from a disjoint pool of THINGS images), 5/5 and 5/5 seeds (original analysis: all pairs, mean-RDM convention). The result is, however, strongly dependent on the evaluation

resolution, which this paper holds fixed at 224px; see arXiv:2608.12408.

**Few reliable differences at higher areas.** At LOC, BP, PC and STDP exceed the random baseline, FA is borderline, and the differences among trained rules are small and not significant. At IT, no two conditions can be distinguished. This may reflect a genuine property of abstract categorical representations, but is also consistent with a capacity limitation: our small CNN trained on 8,000 CIFAR-10 samples may simply lack the representational power to differentiate at higher layers. Prior work with larger models suggests that BP-trained networks have a systematic IT advantage (Schrimpf et al., 2020); whether this advantage reflects the learning rule or the additional model capacity remains an open question. The low reliability of the IT data (lower bound 0.021, with every condition below it) also leaves little room for differences to be resolved with three subjects; true differences may exist but are not resolvable here.

**FA lowest at V1.** FA produces the lowest alignment at V1, significantly below all other conditions, despite reaching 38.8% test accuracy. This is consistent with random feedback filters disrupting the representational structure that convolutional architecture provides, though we cannot rule out that this reflects an optimization artifact specific to FA in convolutional networks rather than a fundamental incompatibility. After pixel control, FA’s V2 score is  $0.002$   $[-0.005, 0.008]$ ; it was already not distinguishable from zero without pixel control ( $0.004$   $[-0.003, 0.011]$ ). This result may be specific to CNNs, where spatial structure in feature maps interacts poorly with random feedback filters (Lillicrap et al., 2016).

**Dissociation between task performance and brain alignment.** Task accuracy and brain alignment are dissociated in our small-scale setting. At V1, the untrained baseline (10.0% test accuracy) exceeds BP (75.0%), and at IT, conditions with test accuracies between 10.0% and 75.0% cannot be distinguished. This suggests that, at least for small CNNs trained on CIFAR-10, the relevant axis is not “how well does it classify” but “what representational structure does the objective induce at each layer.” Whether this dissociation persists at ImageNet scale, where task-optimized models achieve substantially higher Brain-Score alignment

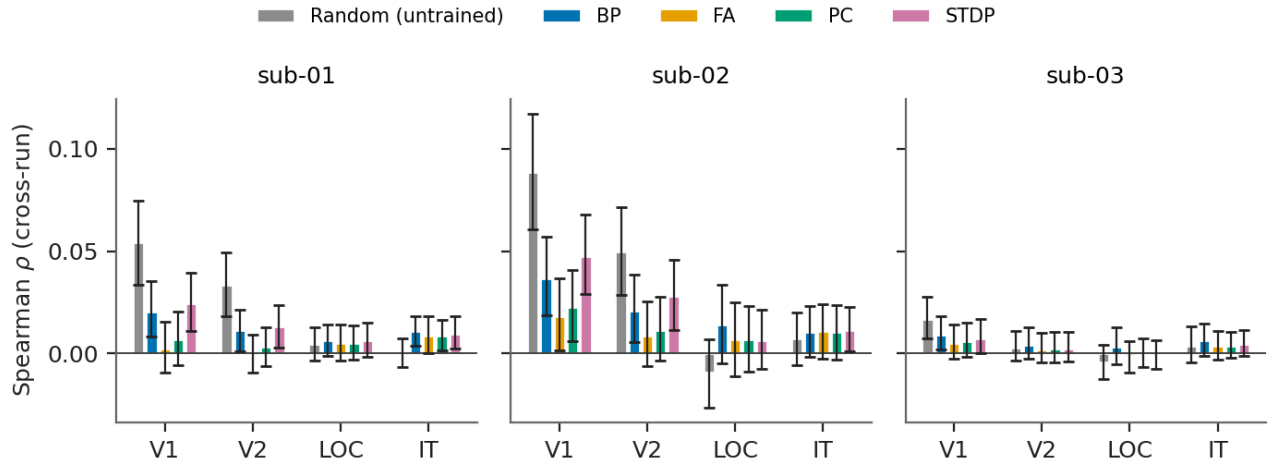


Figure 5: Single-subject RSA scores per learning rule and ROI (cross-run pairs, mean of 5 seeds), with stimulus-bootstrap 95% CIs. Sub-03 shows weaker alignment than sub-01 and sub-02 across conditions and ROIs. In sub-01 and sub-02 the V1 order of the point estimates is that of the group; at LOC, no difference between two trained rules is significant in any single subject.

(Schrimpf et al., 2020), remains an open question.

**Implications.** Our results carry two concrete methodological messages. For computational neuroscience: studies comparing trained models to brain data must always include an untrained baseline with the same architecture. Without this control, architecture effects are confounded with learning effects—our data show this confound can be essentially complete at V1. The same control must be run at the training resolution: the untrained network’s lead over BP at V1 is present at 224 px evaluation but absent at the 32 px training resolution; it arises from the mismatch between training and evaluation resolution, not from training per se degrading early-layer structure (arXiv:2608.12408). A second message comes from the revision itself: when subjects share a presentation order, same-run stimulus pairs can inflate between-subject agreement and the noise ceiling, and should be excluded or modelled.

### 5.1 Limitations

Several limitations bound our conclusions. (1) All models were trained on CIFAR-10 using a small custom CNN, far below the scale at which ResNet-50 or ViT models achieve competitive Brain-Score results. The small architecture limits representational capacity and may contribute to the absence of differences at IT. (2) We have only 3 subjects, with sub-03 showing consistently weak signal; our core findings are driven by sub-01 and sub-02. With three subjects only the lower noise bound is informative. Replication on a larger dataset (e.g., Natural Scenes Dataset,  $N = 8$ ) would strengthen generalizability. (3) STDP was applied to static images via Poisson spike trains, discarding the temporal dy-

namics that STDP was designed to exploit. (4) The best-layer-per-ROI analysis reveals that FC1 sometimes produces higher V1 alignment than Conv1, complicating the anatomical mapping assumption. (5) Gabor-peakedness scores show no significant differences across learning rules, but this metric captures only a coarse proxy for V1-like filter structure. (6) Restricting the analysis to cross-run pairs removes 18.8% of the stimulus pairs; it removes the shared run component, but the sensitivity analyses (appendix) show that the main conclusions do not depend on this choice. (7) We use a single dataset (THINGS-fMRI); cross-dataset replication would strengthen all claims. (8) Models were trained at  $32 \times 32$  resolution but evaluated at  $224 \times 224$ ; the size of the untrained baseline’s V1 advantage depends strongly on the evaluation resolution, which is quantified in arXiv:2608.12408.

## 6 Conclusion

We presented a systematic RSA comparison of four learning rules plus an untrained baseline at  $224 \times 224$  resolution, averaged across 5 seeds, against the THINGS-fMRI benchmark. In our small-scale setup, the learning rule has little effect on cortical alignment: architecture dominates early visual representations (V1/V2), where the untrained baseline exceeds all trained rules and, at V1, is the only condition that reaches the lower noise bound. This advantage holds at the 224 px evaluation resolution; in the model set of the resolution study (same architecture and training, 5 seeds) the V1 gap is  $\Delta\rho = +0.0304$  (95% CI  $[+0.0181, +0.0423]$ ) at 224 px, and when the models are evaluated at their 32 px training resolution it vanishes ( $\Delta\rho = -0.0007$ , 95% CI  $[-0.0109, +0.0090]$ ):

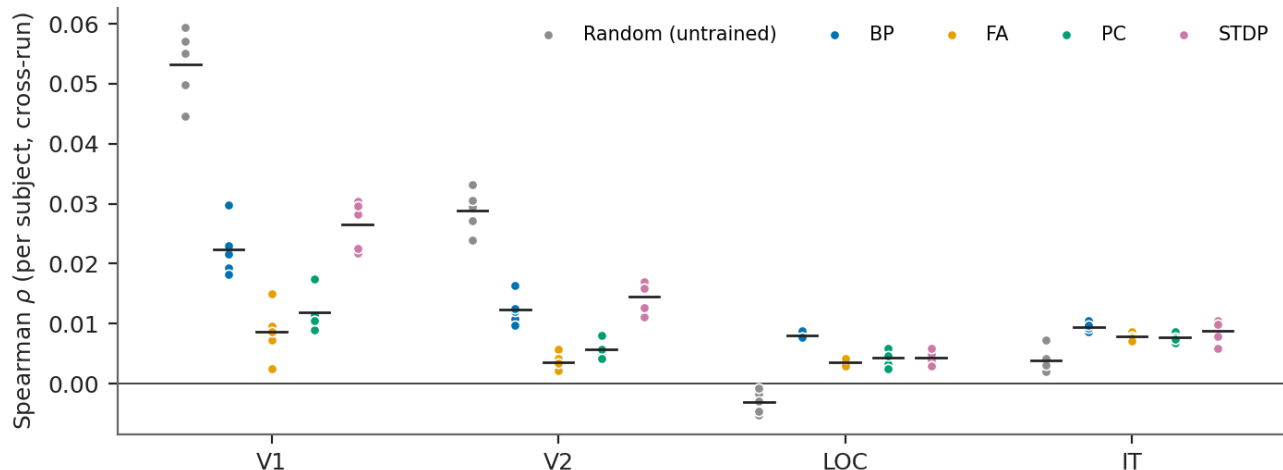


Figure 6: RSA scores of the 5 seeds per condition (dots) and their mean (line),  $\rho_{\text{subj}}$ , cross-run pairs. At V1/V2, between-rule differences are large relative to the seed spread. At IT, the seed spread is small, and the lack of significant differences reflects the stimulus-sampling uncertainty.

BP and the untrained network do not differ at V1, and the untrained network’s lead at 224 px arises from the difference between training and evaluation resolution (arXiv:2608.12408). At LOC, three of the four trained rules exceed the untrained baseline (FA borderline) without differing from each other; at IT, no two conditions can be distinguished at the low reliability of the data—though this may also reflect model capacity limitations. FA consistently produces the lowest alignment at V1 despite above-chance accuracy (38.8%). These results suggest that, at least at this scale, learning-rule comparisons at early areas are dominated by the architecture and by the relation between training and evaluation resolution, and should report both. Whether these findings generalize to larger architectures trained on richer datasets remains an important open question.

## Acknowledgements

The author thanks Martin Schrimpf for the arXiv endorsement and helpful feedback, the creators of the THINGS-fMRI dataset for making their data publicly available, and the Brain-Score team for their evaluation infrastructure.

## Code Availability

Code and results: <https://github.com/nilsleut/learning-rules-rsa/tree/arxiv-v4>.

## A Sensitivity analyses

**Pair set and block cleaning.** Table 5 gives the noise bounds and the per-subject model scores for three pair sets: all stimulus pairs (as in earlier versions), cross-run pairs (main analysis), and cross-run pairs with block cleaning, in which, separately for each subject, the mean of every run-pair block is subtracted from that subject’s RDM and from the model RDM it is compared with. Block cleaning is defined only for the per-subject convention: the three subjects have different run structures, so no single cleaning applies to their average. The last column gives the original (pre-repair) model set on cross-run pairs. Restricting to cross-run pairs lowers the lower bound by 10 % at V1, 28 % at V2, 46 % at LOC and 39 % at IT; block cleaning changes it little. The model scores change by at most 0.002 between variants.

**Conclusions under each variant.** The statements of the main text (untrained exceeds BP at V1 and V2; BP exceeds the untrained network at LOC; FA lowest at V1; no significant differences among trained rules at LOC; none at IT) have the same sign and the same significance in all three pair sets, with one exception. In the repaired model set with block cleaning, BP vs. FA at LOC becomes borderline significant: +0.0050 [0.0001, 0.0102], against +0.0044 [−0.0011, 0.0102] on cross-run pairs without cleaning. We treat it as borderline.

**Layer assignment of V2** (Table 6). The fixed mapping assigns V2 to Conv1. With V2 assigned to Conv2 instead (same pair set, resamples and per-subject convention), the untrained network still exceeds BP, and it and STDP still exceed FA; BP falls below FA, and PC is no longer distinguishable from FA. The ordering among FA, BP and PC at V2 therefore depends on the layer

Table 5: Sensitivity: noise bounds and  $\rho_{\text{subj}}$  (mean of 5 seeds) for all pairs, cross-run pairs (main), and cross-run pairs with per-subject block cleaning; repaired model set. Last column: original model set, cross-run pairs. Lower bound with stimulus-bootstrap 95% CI. The upper bound is listed for completeness; it is not informative at  $N = 3$  (Methods).

| ROI | Quantity    | all pairs            | cross-run            | block-cleaned        | cross-run, original set |
|-----|-------------|----------------------|----------------------|----------------------|-------------------------|
| V1  | lower bound | 0.057 [0.045, 0.069] | 0.051 [0.039, 0.064] | 0.054 [0.041, 0.066] | –                       |
|     | upper bound | 0.575                | 0.572                | 0.572                | –                       |
|     | Random      | 0.053                | 0.053                | 0.053                | 0.053                   |
|     | BP          | 0.021                | 0.022                | 0.021                | 0.024                   |
|     | FA          | 0.007                | 0.009                | 0.006                | 0.009                   |
|     | PC          | 0.011                | 0.012                | 0.010                | 0.039                   |
|     | STDP        | 0.025                | 0.027                | 0.024                | 0.045                   |
| V2  | lower bound | 0.033 [0.025, 0.041] | 0.024 [0.016, 0.032] | 0.026 [0.018, 0.034] | –                       |
|     | upper bound | 0.568                | 0.563                | 0.564                | –                       |
|     | Random      | 0.029                | 0.029                | 0.028                | 0.029                   |
|     | BP          | 0.011                | 0.012                | 0.010                | 0.013                   |
|     | FA          | 0.003                | 0.004                | 0.002                | 0.004                   |
|     | PC          | 0.005                | 0.006                | 0.004                | 0.019                   |
|     | STDP        | 0.014                | 0.015                | 0.012                | 0.024                   |
| LOC | lower bound | 0.034 [0.027, 0.042] | 0.018 [0.011, 0.027] | 0.019 [0.011, 0.027] | –                       |
|     | upper bound | 0.566                | 0.558                | 0.559                | –                       |
|     | Random      | –0.002               | –0.003               | –0.005               | –0.003                  |
|     | BP          | 0.007                | 0.008                | 0.008                | 0.008                   |
|     | FA          | 0.003                | 0.004                | 0.003                | 0.004                   |
|     | PC          | 0.004                | 0.004                | 0.003                | 0.003                   |
|     | STDP        | 0.004                | 0.004                | 0.003                | 0.004                   |
| IT  | lower bound | 0.035 [0.028, 0.042] | 0.021 [0.014, 0.028] | 0.021 [0.014, 0.028] | –                       |
|     | upper bound | 0.563                | 0.556                | 0.556                | –                       |
|     | Random      | 0.005                | 0.004                | 0.003                | 0.004                   |
|     | BP          | 0.008                | 0.009                | 0.009                | 0.010                   |
|     | FA          | 0.007                | 0.008                | 0.006                | 0.008                   |
|     | PC          | 0.007                | 0.008                | 0.007                | 0.008                   |
|     | STDP        | 0.008                | 0.009                | 0.008                | 0.007                   |

assignment, whereas Random  $>$  BP holds under both.

Table 6: Primary V2 differences ( $\rho_{\text{subj}}(A) - \rho_{\text{subj}}(B)$ , stimulus-bootstrap 95% CI; cross-run pairs, repaired model set) with V2 assigned to Conv1 (main analysis) or to Conv2.

| Difference at V2 | Conv1→V2 (main)          | Conv2→V2                   | same sign and significance |
|------------------|--------------------------|----------------------------|----------------------------|
| Random – BP      | +0.0165 [0.0072, 0.0261] | +0.0526 [0.0401, 0.0656]   | yes                        |
| Random – FA      | +0.0252 [0.0150, 0.0361] | +0.0313 [0.0212, 0.0413]   | yes                        |
| BP – FA          | +0.0087 [0.0053, 0.0125] | –0.0213 [–0.0274, –0.0148] | no                         |
| PC – FA          | +0.0021 [0.0005, 0.0037] | –0.0005 [–0.0035, 0.0027]  | no                         |
| STDP – FA        | +0.0110 [0.0077, 0.0146] | +0.0097 [0.0047, 0.0152]   | yes                        |

#### Correction for multiple comparisons (Table 7).

Two-sided bootstrap  $p$ -values from the same 1000 resamples,  $p = (2 \min(\#\{d_b \leq 0\}, \#\{d_b \geq 0\}) + 1) / (B + 1)$ , corrected by Benjamini–Hochberg (false discovery rate) and by Holm (family-wise error rate), each at the 5 % level, within the current family of 7 primary comparisons and within the earlier family of 11, which also contained FA against every other condition at V2. All 7 comparisons remain significant under both corrections and both families.

**Diagnostics of the run structure** (Table 8). If the run component were shared beyond same-run pairs, the mean dissimilarity of each pair of runs would agree between sub-01 and sub-02. The Spearman correlation of their 45 run-pair means is small and not significant at any ROI (exact one-sided  $p$  over all 3,628,800 relabellings of sub-02’s runs). With only 45 run pairs this test has little power, so block cleaning is reported as a robustness check rather than ruled out. Block cleaning changes the V1 luminance fit by  $-2\%$  for sub-03, whose run order differs, and by  $-1\%$  (sub-01) and  $-10\%$  (sub-02); the pre-specified stop criterion (a decrease of more than 30 % for sub-01 or sub-02) is not met.

## B Errata of earlier versions

**Accuracy.** Table 11 gives the accuracies printed in earlier versions next to the test accuracies of this version. The earlier values are accuracies on training batches in the last epoch of the original runs, measured with batch normalisation in training mode; for PC and STDP this

Table 7: Primary comparisons with bootstrap  $p$ -values and adjusted  $p$ -values (Benjamini–Hochberg, Holm) within the family of 7 and the earlier family of 11. Per-subject convention, cross-run pairs, repaired model set. The smallest attainable bootstrap  $p$  is  $1/(B + 1)$ .

| Comparison  | ROI | $\Delta\rho_{\text{subj}}$ [95% CI] | $p_{\text{boot}}$ | BH, 7  | BH, 11 | Holm, 7 | Holm, 11 |
|-------------|-----|-------------------------------------|-------------------|--------|--------|---------|----------|
| Random – BP | V1  | +0.0308 [0.0186, 0.0428]            | 0.0010            | 0.0014 | 0.0014 | 0.0070  | 0.0110   |
| Random – BP | V2  | +0.0165 [0.0072, 0.0261]            | 0.0010            | 0.0014 | 0.0014 | 0.0070  | 0.0110   |
| Random – BP | LOC | −0.0110 [−0.0206, −0.0023]          | 0.0190            | 0.0190 | 0.0190 | 0.0190  | 0.0190   |
| Random – FA | V1  | +0.0446 [0.0313, 0.0580]            | 0.0010            | 0.0014 | 0.0014 | 0.0070  | 0.0110   |
| BP – FA     | V1  | +0.0138 [0.0092, 0.0186]            | 0.0010            | 0.0014 | 0.0014 | 0.0070  | 0.0110   |
| FA – PC     | V1  | −0.0033 [−0.0054, −0.0011]          | 0.0030            | 0.0035 | 0.0037 | 0.0070  | 0.0110   |
| FA – STDP   | V1  | −0.0179 [−0.0229, −0.0130]          | 0.0010            | 0.0014 | 0.0014 | 0.0070  | 0.0110   |

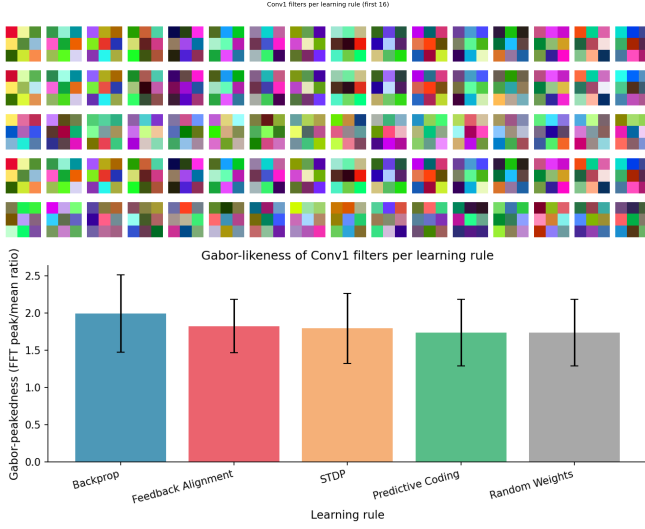


Figure 7: Conv1 filter analysis for all five conditions (original model set; weight-based, so not affected by the evaluation defect). (a) First 16 Conv1 filters per learning rule. All conditions show color-opponent patch structure rather than oriented Gabor patterns, consistent with CIFAR-10 training at  $32 \times 32$ . (b) Gabor-peakedness (FFT spectral peak-to-mean ratio): BP  $1.99 \pm 0.52$ , FA  $1.82 \pm 0.36$ , STDP  $1.79 \pm 0.47$ , PC  $1.73 \pm 0.45$ , Random  $1.73 \pm 0.45$ . Differences are small relative to within-rule variability.

is also the mode in which the evaluation defect left them. They are not test accuracies and are not comparable to Table 1.

Table 12 lists values printed in v2 next to the value its own source gives. The first four are double-rounding errors (rounding first to four, then to three decimals). The V1 gap printed in v2 (+0.044) is that of the resolution study (arXiv:2608.12408, +0.044), while the table it accompanied gives +0.042. The noise bounds printed in v2 (upper 0.11, 0.09, 0.06, 0.07; lower 0.07, 0.05, 0.03, 0.04) could not be traced to any stored result and are replaced by the bounds above.

Table 8: Diagnostics of the run structure. Top: agreement of the run-pair mean dissimilarities between sub-01 and sub-02 (identical presentation order). Bottom: V1 luminance fit per subject on cross-run pairs before and after block cleaning.

| ROI     | $\rho$ (sub-01 vs sub-02, 45 run-pair means)                     | $p$ (exact, one-sided) |
|---------|------------------------------------------------------------------|------------------------|
| V1      | 0.055                                                            | 0.336                  |
| V2      | 0.065                                                            | 0.323                  |
| LOC     | 0.171                                                            | 0.160                  |
| IT      | 0.097                                                            | 0.226                  |
| Subject | V1 luminance fit: cross-run $\rightarrow$ block-cleaned (change) |                        |
| sub-01  | 0.054 $\rightarrow$ 0.054 (−1 %)                                 |                        |
| sub-02  | 0.096 $\rightarrow$ 0.086 (−10 %)                                |                        |
| sub-03  | 0.011 $\rightarrow$ 0.011 (−2 %)                                 |                        |

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Table 9: RSA scores in the mean-RDM convention  $\rho_{\text{mean}}$  (model RDM vs. the average of the three subject RDMs; the convention of earlier versions), with stimulus-bootstrap 95% CI; cross-run pairs, repaired model set, mean of 5 seeds. Bold, <sup>†</sup>, <sup>‡</sup> as in Table 2. The lower bound is defined per subject and is given in Table 2.

| ROI | Layer | lower bound | Random                    | BP                                     | FA                               | PC                               | STDP                             |
|-----|-------|-------------|---------------------------|----------------------------------------|----------------------------------|----------------------------------|----------------------------------|
| V1  | Conv1 | –           | <b>0.077</b> [.055, .099] | 0.034 [.020, .050] <sup>†</sup>        | 0.014 [.000, .028] <sup>†</sup>  | 0.018 [.006, .034] <sup>†</sup>  | 0.039 [.025, .056] <sup>†</sup>  |
| V2  | Conv1 | –           | <b>0.043</b> [.027, .061] | 0.019 [.007, .031] <sup>†</sup>        | 0.005 [–.006, .017] <sup>†</sup> | 0.009 [–.003, .020] <sup>†</sup> | 0.022 [.010, .035] <sup>†</sup>  |
| LOC | Conv3 | –           | –0.007 [–.018, .004]      | <b>0.013</b> [.001, .027] <sup>†</sup> | 0.006 [–.006, .019] <sup>†</sup> | 0.007 [–.004, .018] <sup>†</sup> | 0.007 [–.004, .017] <sup>†</sup> |
| IT  | FC1   | –           | 0.006 [–.003, .016]       | 0.015 [.006, .024]                     | 0.014 [.004, .024]               | 0.013 [.003, .023]               | 0.015 [.005, .024] <sup>‡</sup>  |

Table 10: All pairwise differences  $\rho_{\text{subj}}(A) - \rho_{\text{subj}}(B)$  with stimulus-bootstrap 95% CI (cross-run pairs, repaired model set). Bold: CI excludes 0. <sup>P</sup>: primary comparison (Methods); all others are exploratory, without correction for multiple testing.

| Pair (A – B)  | V1                                              | V2                                            | LOC                                             | IT                        |
|---------------|-------------------------------------------------|-----------------------------------------------|-------------------------------------------------|---------------------------|
| Random – BP   | + <b>0.0308</b> [0.0186, 0.0428] <sup>P</sup>   | + <b>0.0165</b> [0.0072, 0.0261] <sup>P</sup> | – <b>0.0110</b> [–0.0206, –0.0023] <sup>P</sup> | –0.0054 [–0.0144, 0.0036] |
| Random – FA   | + <b>0.0446</b> [0.0313, 0.0580] <sup>P</sup>   | + <b>0.0252</b> [0.0150, 0.0361]              | –0.0066 [–0.0140, 0.0002]                       | –0.0039 [–0.0118, 0.0039] |
| Random – PC   | + <b>0.0413</b> [0.0292, 0.0533]                | + <b>0.0231</b> [0.0140, 0.0327]              | – <b>0.0073</b> [–0.0137, –0.0013]              | –0.0037 [–0.0115, 0.0041] |
| Random – STDP | + <b>0.0267</b> [0.0154, 0.0381]                | + <b>0.0142</b> [0.0051, 0.0238]              | – <b>0.0073</b> [–0.0138, –0.0010]              | –0.0048 [–0.0123, 0.0028] |
| BP – FA       | + <b>0.0138</b> [0.0092, 0.0186] <sup>P</sup>   | + <b>0.0087</b> [0.0053, 0.0125]              | +0.0044 [–0.0011, 0.0102]                       | +0.0016 [–0.0041, 0.0075] |
| BP – PC       | + <b>0.0104</b> [0.0065, 0.0149]                | + <b>0.0066</b> [0.0035, 0.0101]              | +0.0037 [–0.0017, 0.0097]                       | +0.0017 [–0.0036, 0.0072] |
| BP – STDP     | – <b>0.0042</b> [–0.0079, –0.0004]              | –0.0023 [–0.0050, 0.0007]                     | +0.0037 [–0.0017, 0.0097]                       | +0.0006 [–0.0043, 0.0056] |
| FA – PC       | – <b>0.0033</b> [–0.0054, –0.0011] <sup>P</sup> | – <b>0.0021</b> [–0.0037, –0.0005]            | –0.0007 [–0.0032, 0.0018]                       | +0.0001 [–0.0041, 0.0044] |
| FA – STDP     | – <b>0.0179</b> [–0.0229, –0.0130] <sup>P</sup> | – <b>0.0110</b> [–0.0146, –0.0077]            | –0.0007 [–0.0042, 0.0030]                       | –0.0009 [–0.0050, 0.0029] |
| PC – STDP     | – <b>0.0146</b> [–0.0186, –0.0105]              | – <b>0.0089</b> [–0.0119, –0.0062]            | +0.0000 [–0.0030, 0.0030]                       | –0.0011 [–0.0045, 0.0022] |

Table 11: Accuracy printed in v2 (training batches, original runs) and CIFAR-10 test accuracy of the repaired models (seed 0).

| Condition          | v2: training accuracy (%) | v4: test accuracy (%) |
|--------------------|---------------------------|-----------------------|
| Backprop           | 82.4                      | 75.0                  |
| STDP               | 63.2                      | 57.4                  |
| Predictive Coding  | 56.6                      | 56.7                  |
| Feedback Alignment | 39.0                      | 38.8                  |
| Random Weights     | 10.0                      | 10.0                  |

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Table 12: Errata of v2: printed value and the value of its source (original analysis).

| Location in v2             | Quantity                          | printed        | correct     |
|----------------------------|-----------------------------------|----------------|-------------|
| Abstract, §4.2, Tab. 2, §5 | Random, V1                        | 0.076          | 0.075       |
| Abstract, §4.2, Tab. 2, §5 | BP, V1                            | 0.034          | 0.033       |
| Tab. 2                     | BP, V2                            | 0.019          | 0.018       |
| Tab. 2                     | BP, V2, CI upper                  | .023           | .022        |
| Abstract, §1, §4.2, §5     | $\Delta\rho$ Random – BP, V1      | +0.044         | +0.042      |
| §4.8                       | Gabor peakedness, BP              | 2.00           | 1.99        |
| §4.5                       | seed SD range, V1/V2              | 0.003–0.007    | 0.003–0.011 |
| §4.4                       | FDR-significant comparisons of 40 | 30             | 26          |
| §4.4                       | largest FDR $p$ at V1/V2          | $p \leq 0.014$ | 0.0144      |